SPE-229985-MS

Unlocking Reservoir Potential while Minimizing the Footprint: Deployment of Long (2000m) Horizontal Wells to Develop Ultra-Thin, Low-Permeability Carbonate Reservoirs with Reduced Wells Density (A Case Study from Block-5 in Oman)

I. Al Ajmi, Z. Al Kindi, M. Al Rabaani, I. Al Mahrouqi, S. Al Alawi, A. Al Rubkhi, and A. Al Abri, Daleel Petroleum Company

 DOI 10.2118/229985-MS

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

This paper presents a case on the innovative development of an ultra-thin, tight carbonate reservoir through the drilling of extended 2000-meter horizontal wells in an oil field located in Block-5, Oman. This approach resulted in lower wells density compared to the conventional 1000-meter length with minimal operational consequences which effectively reduced carbon emissions and minimized the environmental footprint while unlocking tight reservoirs and improving Net Present Value. The carbonate reservoir in this field is characterized by a thickness of less than two meters and low permeability (less than one millidarcy), leading to substantial production decline due to the restricted flow of formation fluids, which adversely affects overall reservoir productivity. The adopted strategy not only improved productivity but also reduced well density, thereby mitigating the environmental and safety challenges typically associated with high well-density areas.

Between 2021 and 2024, six new long wells were successfully drilled, marking a significant advancement in the field's development. The production results of these long wells were evaluated and compared to the performance of the 1000-meter wells. The extended 2000-meter wells exhibited enhanced productivity, ranging between 10 – 30%. Such successful outcomes, combined with the projected reduction in overall development costs, prompted a comprehensive reassessment of the field development, leading to updates in both the static and dynamic models. Based on the forecasts derived from the updated model, the decision was made to expand the development plan, proposing 11 additional wells in the field. This strategic decision would maximize the recovery of this field while minimizing the well density, compared to the development with 1000-meter wells.

A detailed study was undertaken to establish a clear development strategy for the field. The analysis revealed that extending the horizontal wells to 2000-meter could significantly improve NPV and profitability, driven by the increased productivity of longer wells and the reduction in overall operational costs. Furthermore, the lower number of wells contributes to a reduced environmental footprint, enhancing the sustainability of this field development.

The results observed in this field highlight the potential for implementing the extended lateral wells approach in two other nearby fields with similar geological properties, specifically low permeability and ultra-thin reservoirs. Overall, this innovative strategy would unlock previously untapped reserves, demonstrating the importance of innovation and adaptive strategies in the oil and gas industry. By challenging conventional methods and embracing new approaches, companies can achieve sustainable growth and maximize the value of their assets more efficiently while minimizing their operational and environmental footprint.

Introduction

The targeted reservoir in this field in Block-5, Oman is a very tight carbonate formation, characterized by its low porosity, poor permeability, and thin pay zone. Such formations are common in the region and represent a significant development challenge due to their limited ability to produce hydrocarbons under natural reservoir energy. In this targeted reservoir, fluid movement through the rock matrix is highly restricted as a result of the low permeability, and production rates are typically low unless aided by artificial stimulation or extended reservoir contact. The formation also lacks meaningful natural fracturing or strong aquifer support, further limiting pressure maintenance and production sustainability.

Initial development of the field relied on horizontal wells with lateral lengths of around 1000 meters. These wells were drilled carefully to stay within the thin pay zone and were completed using standard stimulation techniques. While they were effective in achieving early production, performance often declined sharply within the first year. This steep decline was largely due to the small effective drainage area of each well and the inability to maintain consistent pressure support across the wellbore. As a result, field productivity could only be sustained by drilling a large number of wells. However, the need for high well density introduced several operational and economic concerns, which not only increased the overall capital cost of development but also raised environmental and logistical risks. The dense well layout added complexity to field operations, planning, and safety management. These limitations highlighted the need for a more efficient development strategy, one that could achieve broader reservoir contact and better long-term recovery with fewer wells, reduced surface impact, and improved economic returns.

Reservoir Characteristics

Beside poor primary porosity and minimal fracturing, the average net pay in the targeted reservoir in this field is approximately 1.5 to 1.8 meters, with local variations based on sedimentary facies. Formation porosity averages around 15 percent, while permeability is generally in the tight range, with less than 1 millidarcy. Due to its limited vertical thickness and lateral heterogeneity, effective development requires precise well placement and stimulation to ensure connectivity to the reservoir. Additionally, the effective pay zones are difficult to distinguish using conventional logs alone due to subtle porosity variations and the absence of strong resistivity contrast. This makes real-time geo-steering essential during drilling, especially in long laterals, where small structural or sedimentary shifts can result in unintentional exit from the reservoir. Besides that, lateral heterogeneity within the carbonate system introduces variable stimulation response, reinforcing the need for a well-specific completion.

The reservoir also lacks lateral pressure support, making each well responsible for draining a discrete volume. As such, maximizing lateral length is critical to increasing the drainage area per well. This understanding formed the basis for transitioning to extended horizontal sections, which provide higher contact to the productive rock, improve pressure communication, and support longer-term production sustainability.

Revised Development Plan

As part of the business planning cycle in 2021, a comprehensive review of the field development strategy was undertaken. The objective was to assess whether lateral extension could improve field performance while reducing development intensity. The analysis included a reassessment of geological models, integration of recent production data, and full-field dynamic simulation. The outcome of the study indicated that increasing the lateral length to 2000 meters could significantly improve net present value by increasing productivity and reducing the total number of required wells. The simulation results show that long horizontal wells with 2000-meter lateral sections can improve overall recovery by approximately 30% compared to conventional 1000-meter wells (Figure 1). This improvement is primarily driven by the increased reservoir contact area. In the simulation model, it was assumed that the full 2000-meter length would be effectively placed within the reservoir, maximizing drainage and maintaining exposure to productive rock. However, in real field conditions, this ideal scenario is rarely achieved.

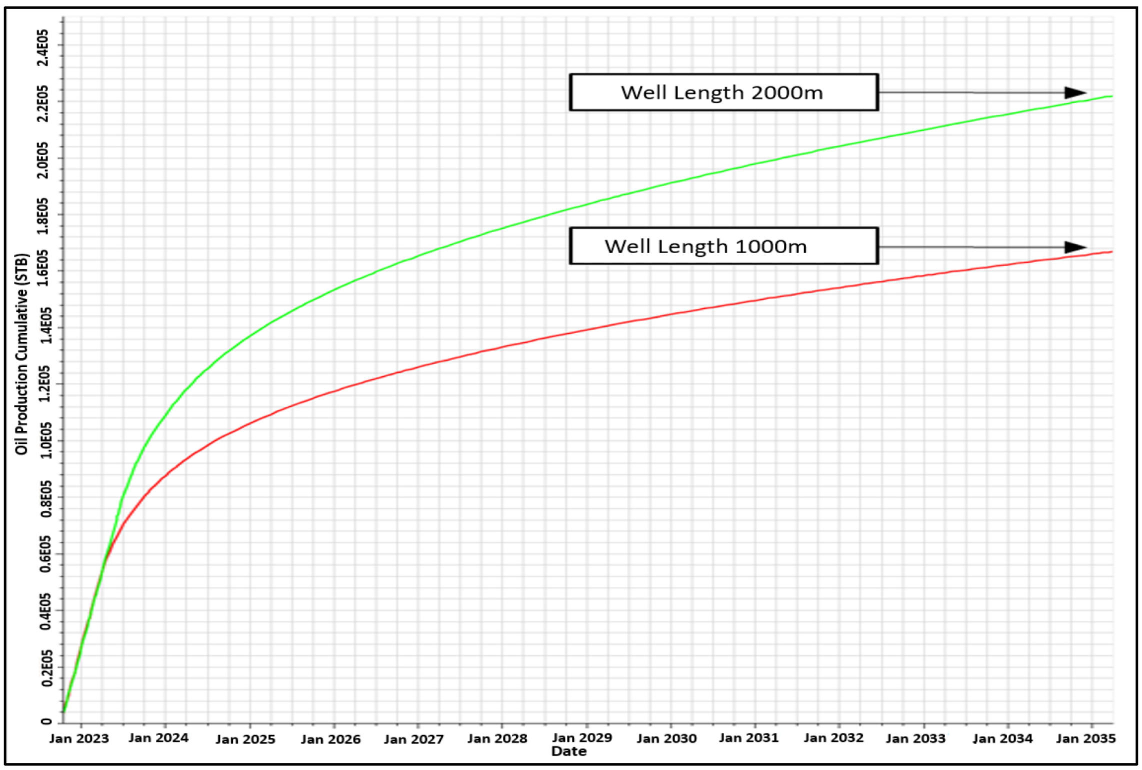


Figure 1—Dynamic model results, comparing cumulative production of two lateral lengths.

The team initiated a pilot phase involving six 2000-meter horizontal wells across different sectors of the reservoir. These wells were carefully designed to maintain placement within the thin pay zone, overcome operational limitations associated with long laterals, and ensure effective stimulation across the full wellbore. The performance of these wells was benchmarked against 1000-meter wells drilled in similar conditions. Data from the pilot wells indicated that actual drilled laterals frequently fell short of the planned 2000 meters target. Due to the extremely thin net pay, real-time geo-steering faced considerable challenges in maintaining precise wellbore placement. Factors such as structural dips, lateral facies variation, and subtle sub-seismic faulting contributed to early exits from the pay zone. Furthermore, even within the drilled lateral, not all sections were within high-quality reservoir rock, meaning the effective reservoir contact is often lower than the drilled length. These geo-steering limitations introduce variability in well performance. Despite these setbacks, the actual field performance of the long wells still showed marked improvements over their 1000-meter counterparts, largely due to the overall increase in reservoir contact. The deviation between simulated and actual lateral lengths also highlights the need to calibrate development models with real well data to ensure realistic forecasting and robust decision-making. The results confirmed the technical and economic advantages of the extended-reach wells and justified the shift in development strategy. The longer wells consistently delivered higher early-time production rates and sustained output over a longer period, which translated into improved Estimated Ultimate Recovery (EUR) per well.

Consequently, the overall economics of this field development improved significantly since the cost per recovered barrel decreased. Economic analysis from the updated dynamic model demonstrated that the same production target could be achieved with approximately 30 - 40% fewer wells. This reduction in well count led to substantial savings not only in drilling and completion costs but also in surface infrastructure, such as flowlines and well pads. In addition, fewer wells resulted in a reduced number of rig moves, minimized site preparations, and lower operational manpower requirements. These indirect savings further improved the Net Present Value (NPV) of the project. As illustrated in Table 1, the 2000-meter well scenario produced more than three times the NPV of the 1000-meter base case, despite higher per-well costs.

Table 1—Net Present Value (NPV) of two field development scenarios.

Scenario	New Wells	RF%*	NPV
2000m long wells	19	19	3.40
Conventional 1000m wells	28	18	1.08

Well Performance and Comparison with 1000-meter Analogs

The six 2000-meter horizontal wells were drilled and completed in different areas of the reservoir, with varying levels of reservoir quality and structural complexity. Each well was compared with at least one offset 1000-meter well in a similar geological setting. The production results showed consistent improvement in early oil rates, pressure stability, and cumulative recovery.

For example, in area A, Well A-2 (2000-meter long) demonstrated a 21 percent higher initial rate compared to Well A-1 (1000-meter long). In area B, Well B-2 (2000-meter long) showed a 30 percent production increase over Well B-1 (1000-meter long) during the first six months of production (Figure 2). In area C, Well C-2 outperformed Well C-1 by 13 percent. These results validated the model predictions and supported the decision to continue with the long-well approach. Some wells were planned as 2000-meter long but were limited to 1200 – 1400 meters lateral sections due to geological constraints such as formation pinch-out or intersection with sub-seismic faults.

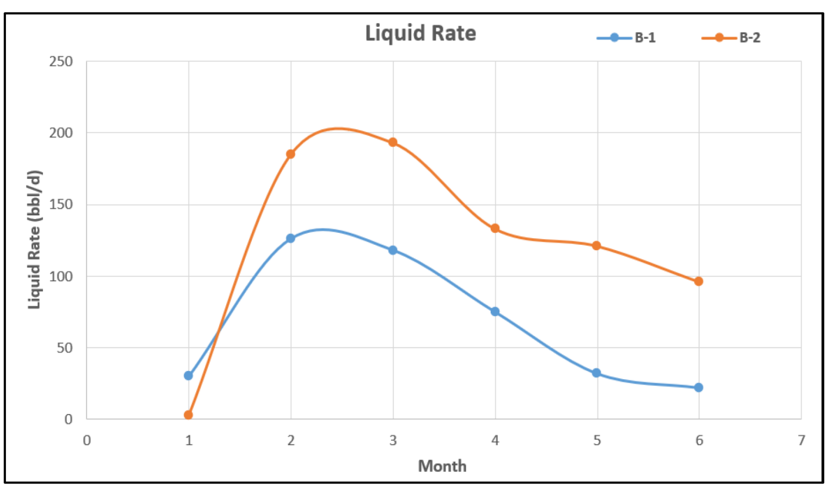
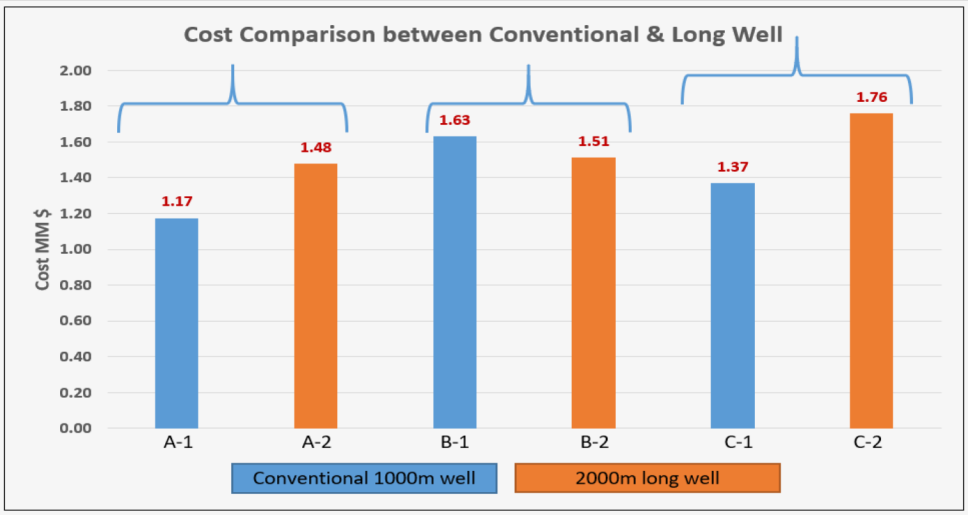


Figure 2—Actual production performance of Well B-2 (2000m long) & Well B-1 (1000m long).

From cost standpoint, the 2000-meter horizontal wells required approximately 10% to 20% more capital expenditure per well compared to the conventional 1000-meter wells (Figure 3). This increase was expected and primarily resulted from the additional time needed to drill the longer laterals, the use of more advanced bottom-hole assemblies to maintain directional control, and higher consumption of drilling fluids and materials. Longer wells also necessitated longer liners and more complex stimulation designs to ensure effective coverage across the extended reservoir contact. These elements collectively contributed to the higher upfront cost. Despite the added investment, the longer wells consistently demonstrated superior performance in terms of both early production rates and overall recovery. The extended contact with the reservoir allowed for greater drainage area per well, reducing the number of wells needed to achieve the same production targets. Additionally, the improved production stability from the long wells led to more predictable cash flow and better utilization of surface facilities, further enhancing project efficiency.



sum cost of conventional wells	sum cost of long wells	Difference
4.17 MM $	4.75 MM $	12%

Figure 3—Capital expenditure per well (1000m vs 2000m).

From a full-cycle economic perspective, the gains in productivity and longer well life more than offset the higher capital expenditure. The lower cost per recovered barrel resulted in higher margins and a stronger project return. As demonstrated earlier in Table 1, the updated model showed a significant improvement in Net Present Value (NPV), with the long well scenario delivering over three times the value of the short well base case. These financial advantages were achieved without compromising technical integrity or operational safety. Moreover, the cost efficiency of the long-well strategy becomes even more compelling when viewed in the context of field-wide development. By drilling fewer, more productive wells, the operator was able to reduce the environmental footprint, accelerate development timelines, and limit long-term operational complexity. This outcome illustrates the broader strategic benefit of designing well architecture not just for immediate cost control, but for long-term value creation and development sustainability.

Operational Execution

Drilling 2000-meter laterals in a reservoir with thickness of less than two meters of net pay (Figure 4) introduced several technical difficulties that required careful planning. One of the main challenges was the high drag & torque on the drilling string, especially near the end of the long horizontal section. This made it difficult to reach the planned total depth (TD) and increased the risk of getting stuck and other down-hole issues related to drilling string. To manage this, the team introduce different ideas such as two different bottom hole assemblies (BHAs), used along with optimizing drilling and hole cleaning practices. The first half of the lateral was drilled using a rotary steerable system to stay within the reservoir. The second half used a vortex motorize BHA, which helped reduce torque, improve hole cleaning, and keep the string moving smoothly. Besides that, a lubricant was added to mud & proper well cleaning was done regularly by reaming down each stand. Mud weight and flow rates were carefully controlled to avoid losses and keep the wellbore stable. Real-time data while drilling was used to monitor gamma readings, formation cuttings & other drilling dynamics. If a sudden change was seen, it was treated as a warning that the well might have exited the reservoir, and immediate adjustments were made.

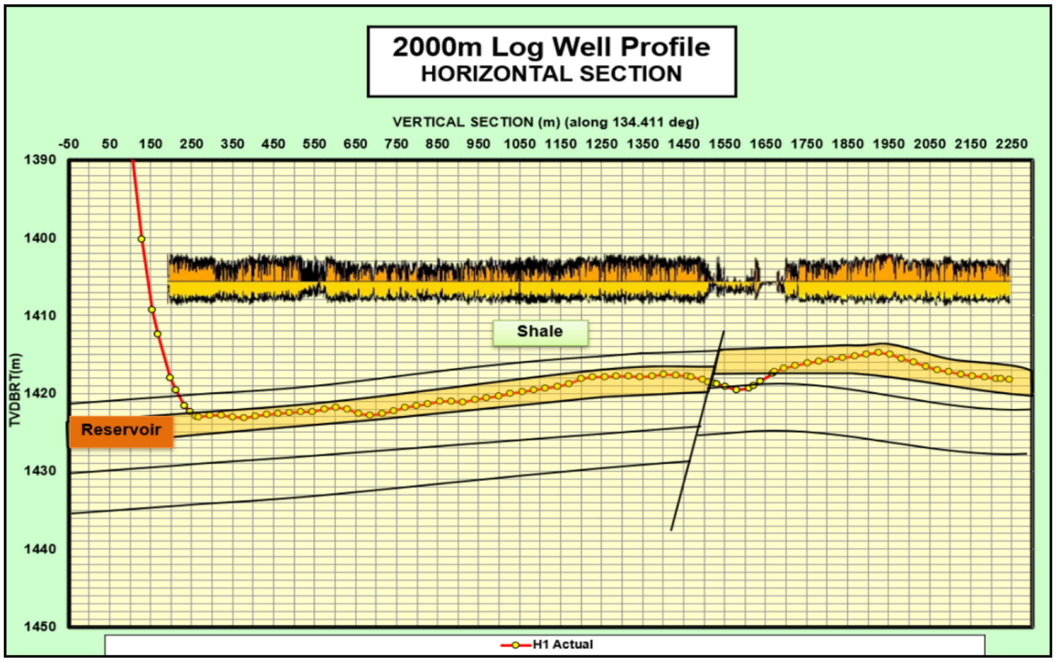


Figure 4—2000m horizontal well profile.

There were also risks of getting stuck or reaching torque limits due to friction along the long horizontal section. To reduce these risks, the drill string was kept moving and the mud weight was maintained at safe levels. Recording formation pressure was also harder in these long wells. Therefore, the number of pressure points recorded was reduced and only the most important depths were selected. Well stimulation posed another challenge. In some wells, it was not possible to reach the full lateral length with coil tubing due to high friction. To improve access, a 4.5-inch slotted liner was run across at least half of the horizontal section (Figure 5). This helped reduce friction and allowed stimulation fluids to be pumped more effectively.

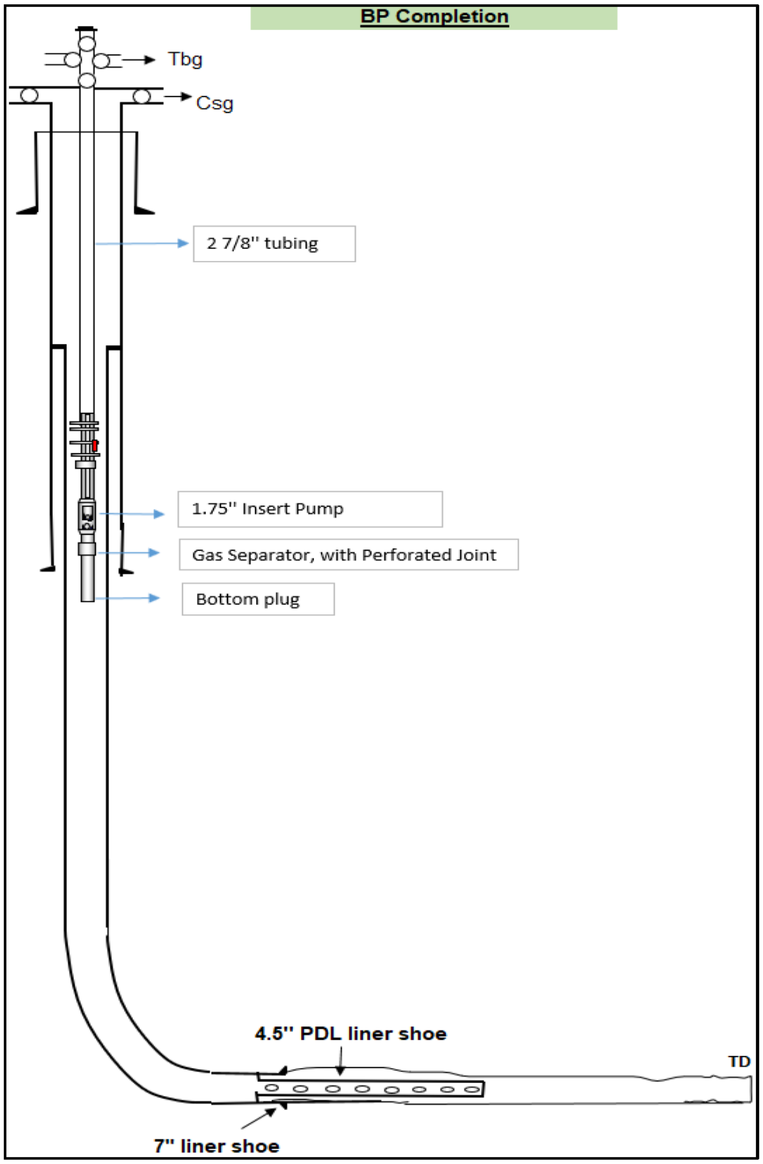


Figure 5—Completion schematic of 2000m horizontal well.

Despite the technical challenges, the team successfully drilled and completed all the long horizontal wells by selecting the right tools, closely monitoring well conditions, and making timely adjustments during operations. These actions were essential to the success of the long-well approach in such a thin and tight reservoir.

Execution time and costs were effectively controlled through detailed planning and strong coordination between the drilling, geology, and production teams. Although longer wells introduced additional complexity, drilling durations remained within acceptable limits. Lessons learned from each well were used to improve the performance of the next, particularly in optimizing BHA configurations, improving wellbore cleaning, and enhancing stimulation methods. This continuous learning process contributed to consistent execution and reliable results across the campaign.

Field Development Impact and Sustainability Benefits

The positive outcomes from the initial batch of 2000-meter horizontal wells played a critical role in revising the field's production forecast and resource outlook. The improved production performance, better pressure response, and increased recovery from the long wells provided solid evidence that the reservoir could deliver sustained output when developed with extended laterals. These field results provided necessary data to update the reservoir simulation model, which resulted in better calibration of recovery factors and well performance forecasts. As a result of this improved understanding, approximately 1.6 million barrels of oil previously classified as contingent resources were reclassified as proven reserves. This reclassification marked an important milestone for the field, as it reflected both increased technical confidence in the reservoir and greater certainty that the remaining resources could be economically produced under this revised development plan. The upgrade was not simply a modeling adjustment, it was backed by physical well data, pressure trends, and production curves that closely matched forecasted behavior, confirming that the long horizontal wells could reliably unlock reserves in even the thinnest sections of the reservoir.

With the updated model, the field development plan was revised to include 11 additional long horizontal wells. These wells were planned using the same geo-steering strategy, drilling assembly setup, and completion design that had proven successful in the pilot campaign. The consistency of results across structurally and geologically varied parts of the reservoir gave further confidence that this approach could be scaled field-wide. This updated plan not only increased economic value of the field but also extended its production life, secured additional investments, and improved long-term planning around infrastructure and facility capacity.

In essence, the reclassification of reserves was both a technical and strategic turning point, signaling a shift from cautious resource testing to full-scale development. More importantly, the success of this approach created a clear development strategy that can now be applied to other assets with similar geological characteristics. Several nearby fields exhibit the same challenges, ultra-thin carbonate pay zones, low permeability, and limited natural drive. This project offers a validated model for how to approach these reservoirs using long horizontal wells, precise geo-steering, and optimized completion techniques.

Conclusion

The deployment of 2000-meter horizontal wells in Block-5 has demonstrated that extended-reach drilling is not only technically viable but also economically advantageous in the development of ultra-thin, low-permeability carbonate reservoirs.

- The long laterals significantly enhanced reservoir contact, compared to conventional 1000-meter wells, resulting in better reservoir drainage, more stable pressure behavior, and improved recovery per well.
- This approach leads to in a substantial reduction in the number of wells needed to achieve production targets. This directly reduced capital and operational expenditures while also lowering the environmental footprint of the development. With fewer wells and less surface infrastructure, the project achieved better environmental compliance, safer operations, and more sustainable long-term planning.
- The improved economics and recovery potential from the pilot wells also enabled the reclassification of a significant portion of the field's resources from contingent to proven reserves.
- The project overcame the inherent challenges of drilling and completing long laterals in thin reservoirs through tailored wellbore design, precise geo-steering, and optimized stimulation methods.
- Each well contributed valuable insights that helped refine drilling practices and improve the efficiency of subsequent operations.
- This case highlights the value of rethinking traditional development approaches, especially in reservoirs that were once considered uneconomical.

Ultimately, this study reinforces the importance of reservoir-driven well design, data-informed execution, and adaptive field development strategies. The successful application of long horizontal wells in this setting opens the door for similar solutions across a wide range of tight and challenging reservoirs, supporting both improved resource recovery and responsible field development.

References

Alyan, M., Al Shamouty, B., Tiwari, A., Mikula, M., Martin, J., and Al Sowaidi, A. 2018. *Successful Utilization of Extra-Long Lateral MRC Wells in Optimized Development of Tight and Thin Carbonates*. Paper presented at the SPE Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 12–15 November. SPE-193264-MS.

Dosunmu, I., and Osisanya, S. 2015. *An Economic Approach to Horizontal Well Length Optimization*. Paper presented at the SPE Abu Dhabi International Petroleum Exhibition and Conference, Abu Dhabi, UAE, 9–12 November. SPE-177866-MS.

Cho, H. 2001. *Integrated Optimization on Long Horizontal Well Length*. Paper presented at the SPE Hydrocarbon Economics and Evaluation Symposium, Dallas, Texas, USA, 2–3 April. SPE-68599-MS.

Hui, P., Qiquan, R., Yong, L., and Yongjun, W. 2017. *Development Strategy Optimization for Different Kinds of Tight Oil Reservoirs*. Paper presented at the SPE Kingdom of Saudi Arabia Annual Technical Symposium and Exhibition, Dammam, Saudi Arabia, 24–27 April. SPE-188070-MS.